Propensity Score Matching

ECON726

Tara Sullivan

Tara Sullivan
tara.sullivan55@login.cuny.edu

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Housekeeping

Today:

- Discuss your project plan
- Propensity score matching
- Jupyter notebook: link

Homework

- Read MHE 5.1. and 5.2. (Fixed effects and differences-in-differences)
  - We will focus on diff-in-diff next class, but many of your papers use fixed effects
- Submit your project plan. **This is 15% of your final grade**
- By next week, you need to have:
  - Read your paper
  - Verified you can access and run your replication code
  - Submitted your project plan

No class or office hours next week!

If you personally need adjustments to these deadlines let me know in advance

Paper selection: Due ~~**September 16**~~ **September 22** [5% of final grade]

- Done!

Project plan: Due **October 14** [15% of final grade]

- Summarize the public policy or business context of the paper. Write down the specific question the paper is evaluating. Discuss the data your are using (both your replication data, and data for your extension) Discuss the extensions you are considering, and, if necessary, where you will be getting that data.

Replication results: Due ~~**November 4**~~ **November 3**: [10% of final grade]

- You should submit some of your main replication results. The final results should be formatted as tables and charts in a word document or PDF, submitted on Brightspace. Your code and data should accessible on GitHub.

Presentation: Due **December 2** & **December 9** [10% of final grade]

- Prepare a 5-10 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 16** [35% of final grade]

- Your paper should contain 5-8 pages of text, and 5-8 pages of figures (double spaced, 12 point font). You are expected to use python, and must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Paper ideas

Note: you might need to google for these data and code, but they are readily available on the internet.

- Matching estimators paper: Angrist (1998, tables II and V; link; data)
- Propensity Score Estimator: Deheija and Wahba (1999; link; data)
- Synthetic Control
  - German reunification: Abadie, Diamond and Hainmuller (2015; link)
  - Cost of conflict in Basque Country: Abadie and Gardeazabal (2003; link)
  - Smoking legislation in California: Abadie, Diamond, and Hainmuller (2010; link)

Outline

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

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Review: matching

Variables:

- Y_i: Outcome variable
- D_i: Binary treatment dummy variable
- X_i: Covariates

Under conditional independence assumption (CIA) $\{Y_{0i}, Y_{1i}\} \perp D_i | X_i$, we can construct causal matching estimators:

$$\begin{aligned}
\delta_{ToT} &= \mathbf{E}[Y_{1i} - Y_{0i} | D_i = 1] \\
&= \sum_x \delta_x \mathbf{P}(X_i = x | D_i = 1) \\
\delta_{ATE} &= \mathbf{E}[Y_{1i} - Y_{0i}] \\
&= \sum_x \delta x \mathbf{P}(X_i = x)
\end{aligned}$$

where:

$$\delta_X = \mathbf{E}[Y_i | X_i, D_i = 1] - \mathbf{E}[Y_i | X_i, D_i = 0]$$

Note: matching and regression are both weighted averages of covariate specific differences

Motivating example

"Growth mindset" example (heavily borrows from: link)

- If you think your intelligence is fixed, there is no incentive to study; academic outcomes are worse
- If you have a "growth mindset," there are returns to intelligence; your academic outcomes are better

The National Study of Learning Mindsets

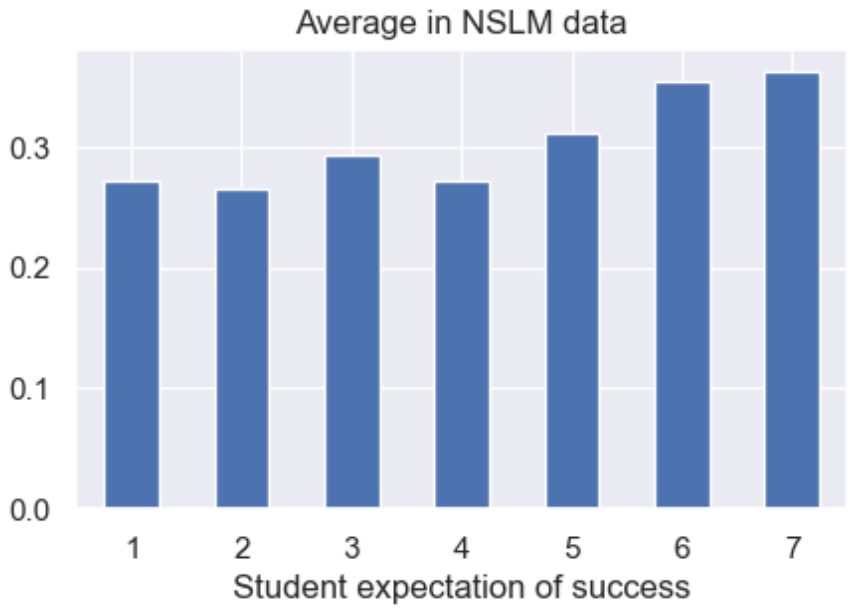
- Evaluate the impact of a nudge-like intervention designed to instill students with a growth mindset on student achievement
- Students randomly given the opportunity to participate in "growth mindset" seminar in high school; academic outcomes in college were then recorded
- Problem: students with the highest expectation of success were more likely to receive the treatment
  - Opportunity to participate in the seminar was random
  - Those who moved forward with the opportunity may have been more ambitious than those who did not

Athey and Wagner (2016) generate simulated data: link

Motivating example (cont)

Variables:

- Y_i: Outcome variable (achievement score in college)
- D_i: Binary treatment dummy variable (participating in "growth mindset") seminar
- X_i: Covariates (individual and school demographics; family characteristics; survey about expectations of future success)



More ambitious students may be more willing to go to the seminar, leading to selection bias. How can we correct for that?

Propensity Score Theorem

Define the propensity score (or balancing score) to be:

$$\begin{aligned} p(X_i) =& \mathbf{E}[D_i|X_i] \\ =& \sum_{d=0}^{1} d \cdot \mathbf{P}(D_i = d|X_i) \\ =& 0 \cdot \mathbf{P}(D_i = 0|X_i) + 1 \cdot \mathbf{P}(D_i = 1|X_i) \\ =& \mathbf{P}(D_i = 1|X_i) \end{aligned}$$

Note: the propensity score equals the probability of treatment for binary treatments

Suppose the CIA holds such that $\{Y_{0i}, Y_{1i}\} \perp D_i|X_i$. Then $\{Y_{0i}, Y_{1i}\} \perp D_i|p(X_i)$

What does this mean?

- You don't need to directly control for confounders X to achieve CIA
- Instead, it is sufficient to control for a single variable, $p(X_i)$

Propensity Score Theorem: Given propensity score:

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Proof:
First, instead of writing $\{Y_{0i}, Y_{1i}\}$, we'll write Y_{ji}, where $j \in 0, 1$.

Next, note that the conditional expectation of the treatment variable equals the conditional probability of treatment (because the treatment is binary):

$$\begin{aligned}
\mathbf{E}[D_i|Y_{ji}, p(X_i)] =& \sum_{d=0}^{1} d\mathbf{P}(D_i = d|Y_{ji}, p(X_i)) \\
=& 0 \cdot \mathbf{P}(D_i = 0|Y_{ji}, p(X_i)) + 1 \cdot \mathbf{P}(D_i = 1|Y_{ji}, p(X_i)) \\
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First consider the left hand side:

$$\mathbf{E}[D_i|X_i, p(X_i)] = \mathbf{E}[D_i|X_i] = p(X_i)$$

In words: $p(X_i)$ is a function of X_i, so conditioning on X_i and $p(X_i)$ does not give any more information than conditioning on X_i alone.

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$$\mathbf{E}[D_i|p(X_i)] = \mathbf{E}\Big[\mathbf{E}[D_i|X_i, p(X_i)]\Big|p(X_i)\Big] \qquad \because \text{Law of iterated expectations}$$

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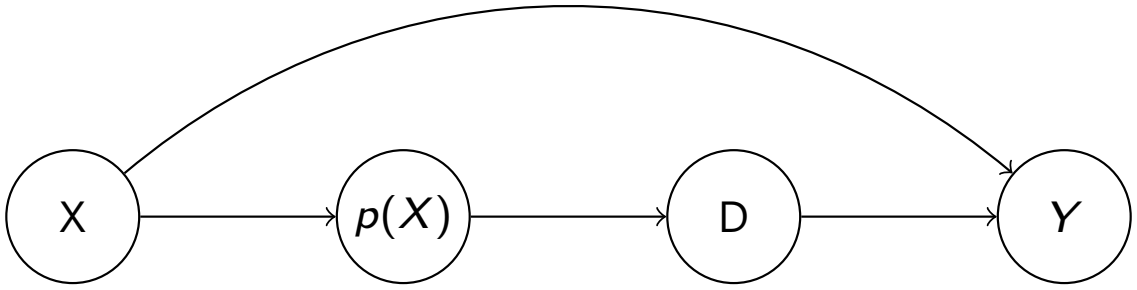
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Because $p(X_i) = p(X_i)$, the statement is true.



- If I know what $p(X)$ is, then X alone doesn't tell me anything else about what the probability of treatment is
- Controlling for $p(X)$ acts the same way as controlling for X directly

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Propensity scores and treatment effects

How are propensity scores used in practice?

- First, estimate the propensity score
  - Use a parametric model, like a logit or probit model
- Next generate the effect of treatment
  - Match on estimated score (direct propensity score matching)
  - OR use weighting scheme described next

Direct propensity score matching

Given a propensity score, $p(X_i)$, and the CIA $\{Y_{0i}, Y_{1i}\} \perp D_i | X_i$:

$$\begin{aligned}\delta_{ToT} =& \mathbf{E}[Y_{1i} - Y_{0i} | D_i = 1] \\ =& \mathbf{E}\left[\mathbf{E}[Y_i | p(X_i), D_i = 1] - \mathbf{E}[Y_i | p(X_i), D_i = 0]\right]\end{aligned}$$

Just like we did for matching, you can

- Stratify the sample in to values of $p(X_i)$
- Find treatment effects conditional on similar $p(X_i)$
- Weigh these treatment effects by the distribution of $p(X_i)$ conditional on treatment

Propensity score weights

Alternative: estimate treatment effects using propensity score weights. Final equation:

$$\mathbf{E}[Y_{1i} - Y_{0i}] = \mathbf{E}\left[\frac{Y_i\left[D_i - p(X_i)\right]}{p(X_i)(1 - p(X_i))}\right]$$

We'll go through the math. Motivation for this method:

- Avoids the matching step
- The concept of inverse probability weighting is important in more advanced topics

To prove this, we'll have to prove a few interim steps

Fact #1:

$$\mathbf{E}\left[\frac{Y_i D_i}{p(X_i)}\right] = \mathbf{E}[Y_{1i}]$$

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Next, recall $Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$:

$$Y_i D_i = (Y_{0i} + (Y_{1i} - Y_{0i})D_i)D_i$$

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Using the above equality, the conditional independence assumption, and the definition of the propensity score:

$$\mathbf{E}[Y_i D_i | X_i] = \mathbf{E}[Y_{1i} D_i | X_i] \qquad \because Y_i D_i = Y_{1i} D_i$$

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&= \mathbf{E}\left[\frac{p(X_i)}{p(X_i)}\mathbf{E}[Y_{1i}|X_i]\right]
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&= \mathbf{E}\left[\frac{1}{p(X_i)}\mathbf{E}[Y_{1i}|X_i]p(x)\right] && \because \mathbf{E}[Y_i D_i|X_i] = \mathbf{E}[Y_{1i}|X_i]p(X_i) \\
&= \mathbf{E}\left[\frac{p(X_i)}{p(X_i)}\mathbf{E}[Y_{1i}|X_i]\right] \\
&= \mathbf{E}[Y_{1i}] && \because \text{Law of iterated expectations}
\end{aligned}$$

Fact #2:

$$\mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right] = \mathbf{E}[Y_{0i}]$$

Proof: Recall $Y_i = Y_{0i} + (Y_{1i} - Y_{0i})$, and note $D_i(1-D_i) = 0$ for $D_i = 0$ and $D_i = 1$:

$$Y_i(1-D_i)$$

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Using the above equality, the conditional independence assumption, and the definition of the propensity score:

$$\mathbf{E}[Y_i(1-D_i)|X_i] = \mathbf{E}\left[\, Y_{0i}(1-D_i)|\, X_i\right]$$

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Proof (cont.):

We've seen:

$$\mathbf{E}[Y_i(1-D_i)|X_i] = \mathbf{E}[Y_{0i}|X_i](1-p(X_i))$$

Now we can estimate $\mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right]$:

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$$\mathbf{E}\left[\frac{Y_i D_i}{p(X_i)}\right] = \mathbf{E}[Y_{1i}], \qquad \mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right] = \mathbf{E}[Y_{0i}]$$

Average treatment effects can be estimated from:

$$\begin{aligned}
\mathbf{E}[Y_{1i} - Y_{0i}] =& \mathbf{E}\left[\frac{Y_i D_i}{p(X_i)} - \frac{Y_i(1-D_i)}{1-p(X_i)}\right] \\
=& \mathbf{E}\left[\frac{Y_i D_i(1-p(X_i)) - Y_i(1-D_i)p(X_i)}{p(X_i)(1-p(X_i))}\right] \\
=& \mathbf{E}\left[\frac{Y_i\left[D_i(1-p(X_i)) - (1-D_i)p(X_i)\right]}{p(X_i)(1-p(X_i))}\right] \\
=& \mathbf{E}\left[\frac{Y_i\left[D_i - D_i p(X_i) - p(X_i) + D_i p(X_i)\right]}{p(X_i)(1-p(X_i))}\right] \\
=& \mathbf{E}\left[\frac{Y_i\left[D_i - p(X_i)\right]}{p(X_i)(1-p(X_i))}\right]
\end{aligned}$$

Horvitz-Thompson Propensity score weighting:

$$\mathbf{E}[Y_{1i} - Y_{0i}] = \mathbf{E}\left[\frac{Y_i\left[D_i - p(X_i)\right]}{p(X_i)(1 - p(X_i))}\right]$$

- Uses propensity score weighting to recover average treatment effects
- Using the propensity score rather than identification by conditioning on X is a useful empirical strategy when X is high-dimensional and $p(X)$ is available (or can be approximated accurately)
- MHE shows how this can be considered similar to the regression estimator

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Inverse Probability of Treatment Weighting

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Consider the first element:

- $Y_i D_i$: Isolates Y_i that are treated
- $p(X_i)$: Inverse probability of being treated
  - Probabilities are between 0 and 1
  - Low probability events $\implies$ inverse probability high

$$p(X) = 0.1 = \frac{1}{10} \implies \frac{1}{p(X)} = \frac{1}{1/10} = 10$$

  - High probability events $\implies$ inverse probability low

$$p(X) = 0.8 = \frac{4}{5} \implies \frac{1}{p(X)} = \frac{1}{4/5} = 1.25$$

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The same reasoning applies to the second element

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The same reasoning applies to the second element

IPTW: valuable in a clinical setting to generate balanced data

Structural model and connection to machine learning

Note: the following is from *Business Data Science* by Matt Taddy, Chapter 6

Consider a common structural model:

$$Y = D\gamma + X'\beta + \epsilon, \quad \epsilon|D, X = 0$$
$$D = X'\tau + v, \quad v|X = 0$$

Note:

- $\epsilon|D, X = 0$ and $v|X = 0$ encode CIA

Two-step **linear treatment effects (LTE)** model

- First estimate treatment D_i
- Then estimate treatment effects γ

Dimensionality starts to matter here

- If n (number of observations) are much greater than p (number of X variables), you can use regression models
- If you don't have $n >> p$, traditional methods break down

Estimation of LTE

Structural model:

$$Y = D\gamma + X'\beta + \epsilon, \quad \epsilon|D, X = 0$$
$$D = X'\tau + v, \quad v|X = 0$$

Estimation procedure:

1. Estimate $\mathbf{E}[D|X] = X'\tau$. Collect fitted values $\hat{D} = X'\hat{\tau}$
2. Estimate:
$$\mathbf{E}[Y|X, D] = \hat{D}\psi + D\gamma + X'\beta$$

The ψ is the treatment effect of D on Y.

LTE and propensity score matching

Structural model in propensity score models:

$$Y = D\gamma + X'\beta + \epsilon, \quad \epsilon|D, X = 0$$
$$D \sim \text{Bernoulli}\left(p(X)\right)$$
$$p(X) = \frac{e^{X'\tau}}{1 + e^{X'\tau}}$$

- D is modeled from a Bernoulli distribution (i.e. coin toss) with probability p
- The probability that $D = 1$ is modeled using a **logistic regression**

Once you estimate the propensity score $p(X)$, you can find treatment effects by:

1. Fitting the linear regression: $\mathbf{E}[Y|D, \hat{p}(X)] = \alpha + D\gamma + \hat{p}(X)\psi$
2. Constructing propensity score weighting estimates: $\mathbf{E}\left[\left.\frac{Y}{p(\hat{X})}\right| D\right] = \alpha + D\gamma$
3. Matching

Hahn (1998) shows there is a cost to propensity score matching in terms of asymptotic efficiency

- What is the advantage of, say, controlling for propensity score instead of controlling for covariates in a regression?

MHE : "An essential element of propensity score methods is the use of prior knowledge for dimension reduction. The statistical payoff is an improvement in finite sample behavior. If you're not prepared to smooth, restrict, or otherwise reduce the dimensionality of the matching problem in a manner that has real empirical consequences, the you might as well go for full covariate matching or saturated regression control."

- This is very prescient! Now, in the era of big data, dimensionality reduction is key
- Propensity scores have a new value in double-robust machine learning; see chapter 9 of `https://chapters.causalml-book.org/CausalML_chap_9.pdf`

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Why logistic regression

- We have to use some estimation procedure to predict propensity scores $p(X)$
- Logistic regression is a common method
- Quick review of logistic regression and generalized linear models

Linear regression

Assume we have data $\{y_i, x_{1i}, x_{2i}, \ldots, x_{ki}\} = \{y_i, \mathbf{x}_i\}$. A linear regression model takes the form:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + \epsilon_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i, \quad i = 1, \ldots, n$$

We can stack these equations for all n equations in the data:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

To estimate this model using least squares, find the coefficient matrix $\boldsymbol{\beta}$ that minimizes the sum of squared residuals:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} = \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{K} x_{ji}\beta_j \right)^2$$

A more math-y way of saying this using norm notation:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} ||\mathbf{y} - \mathbf{X}\boldsymbol{\beta}||^2$$

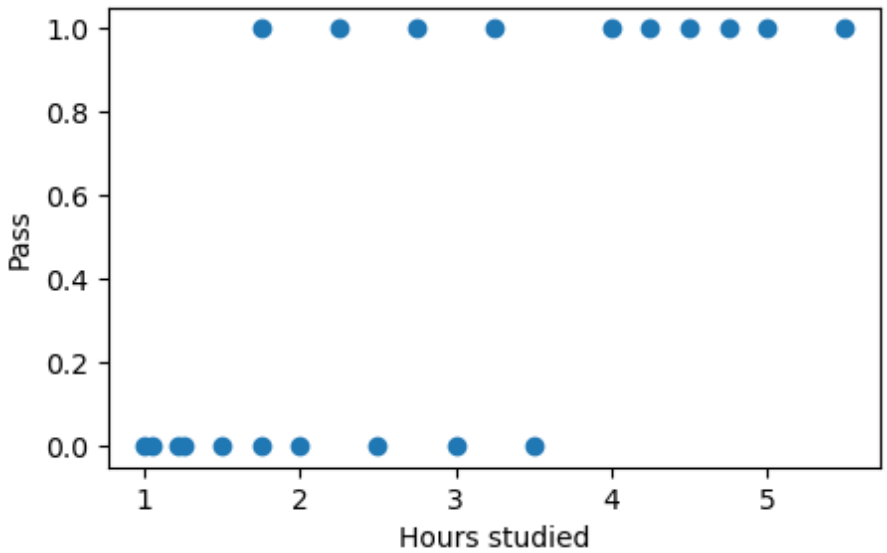
- Just think about the norm notation $||\cdot||$ communicating size in a multidimensional space

Limitations of linear model

We are trying to predict treatment, D_i

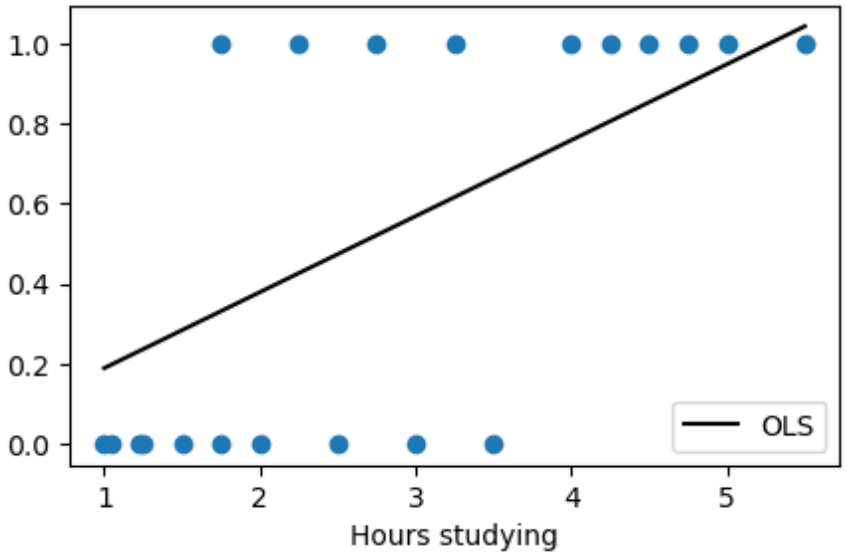
- Always takes on values equal to zero or 1
- Do we really want a linear model?

Consider data on hours studied and passing/failing a test ($D = 1$ implies a student passed)



OLS

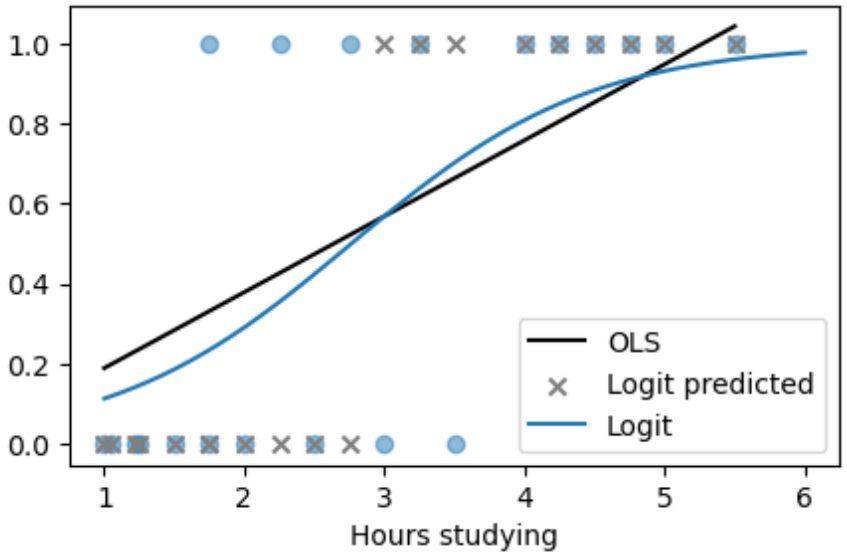
We could estimate using a linear model:



- Line isn't a great fit

Logit

We could estimate using a linear model:



- Logistic line is a better fit
- Less important: output is binary